Equation (4.3.38) is the classical beam deflection formula for a simply supported beam subjected to a concentrated load at mid-span.

Using Eq. (4.3.38), the deflection is obtained as

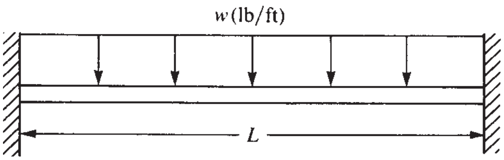
$$v_2 = -2.474 \times 10^{-4} \text{ m} \tag{4.3.39}$$

Comparing the deflections obtained using the shear-correction factor with the deflection predicted using the beam-bending contribution only, we obtain

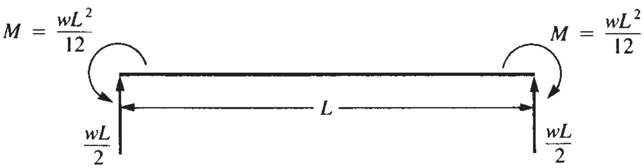
$$\% \text{ change} = \frac{2.597 - 2.474}{2.474} \times 100 = 4.97\% \text{ difference}$$

4.4 Distributed Loading

Beam members can support distributed loading as well as concentrated nodal loading. Therefore, we must be able to account for distributed loading. Consider the fixed-fixed beam subjected to a uniformly distributed loading w shown in Figure 4–21. The reactions, determined from structural analysis theory [2], are shown in Figure 4–22. These reactions are called *fixed-end reactions*. In general, **fixed-end reactions** are those reactions at the ends of an element if the ends of the element are assumed to be fixed—that is, if displacements and rotations are prevented. (Those of you who are unfamiliar with the analysis of indeterminate structures should assume these reactions as given and proceed with the rest of the discussion; we will develop these results in a subsequent presentation of the work-equivalence method.) Therefore, guided by the results from structural analysis for the case of a uniformly distributed load, we replace the load by concentrated nodal forces and moments tending to have the same effect on the beam as the actual distributed load. Figure 4–23 illustrates this idea for a beam. We have replaced the uniformly distributed load by a statically equivalent force system consisting of a concentrated nodal force and moment at each end of the member carrying the distributed load. That is, both the statically equivalent concentrated nodal forces and moments and the original distributed load have the same resultant force and same moment about an arbitrarily chosen point.



■ Figure 4–21 Fixed-fixed beam subjected to a uniformly distributed load



■ Figure 4–22 Fixed-end reactions for the beam of Figure 4–21

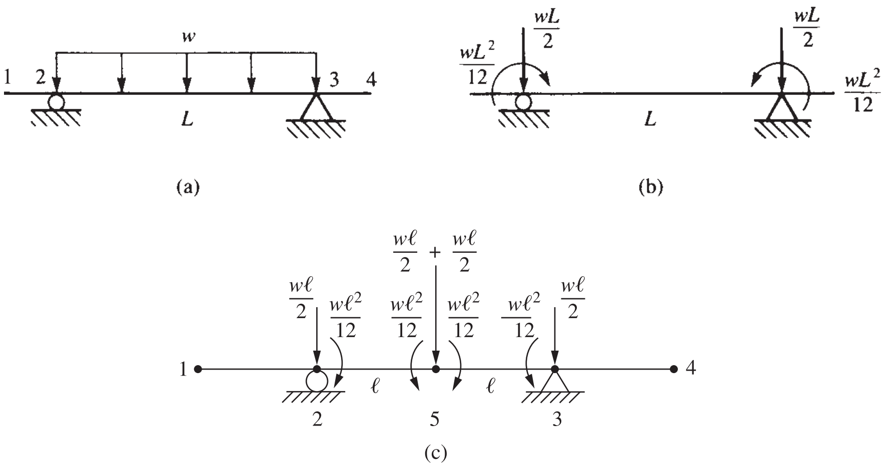


Figure 4–23 (a) Beam with a distributed load, (b) the equivalent nodal force system, and (c) the enlarged beam (for clarity's sake) with equivalent nodal force system when node 5 is added to the midspan

These statically equivalent forces are always of opposite sign from the fixed-end reactions. If we want to analyze the behavior of loaded member 2–3 in better detail, we can place a node at midspan and use the same procedure just described for each of the two elements representing the horizontal member. That is, to determine the maximum deflection and maximum moment in the beam span, a node 5 is needed at midspan of beam segment 2–3, and work-equivalent forces and moments are applied to each element (from node 2 to node 5 and from node 5 to node 3) shown in Figure 4–23(c).

Work-Equivalence Method

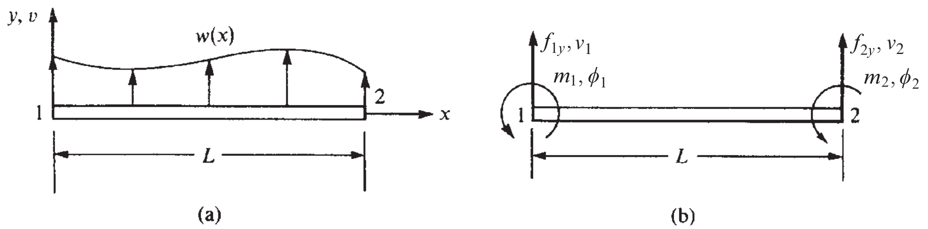
We can use the work-equivalence method to replace a distributed load by a set of discrete loads. This method is based on the concept that the work of the distributed load $w(x)$ in going through the displacement field $v(x)$ is equal to the work done by nodal loads f_{iy} and m_i in going through nodal displacements v_i and ϕ_i for arbitrary nodal displacements. To illustrate the method, we consider the example shown in Figure 4–24. The work due to the distributed load is given by

$$W_{\text{distributed}} = \int_0^L w(x)v(x)\,dx \tag{4.4.1}$$

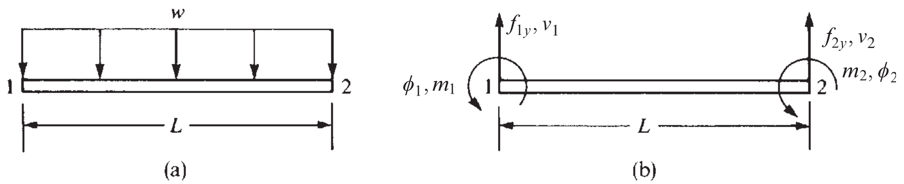
where $v(x)$ is the transverse displacement given by Eq. (4.1.4). The work due to the discrete nodal forces is given by

$$W_{\text{discrete}} = m_1\phi_1 + m_2\phi_2 + f_{1y}v_1 + f_{2y}v_2 \tag{4.4.2}$$

We can then determine the nodal moments and forces m_1, m_2, f_{1y}, and f_{2y} used to replace the distributed load by using the concept of work equivalence—that is, by setting $W_{\text{distributed}} = W_{\text{discrete}}$ for arbitrary displacements ϕ_1, ϕ_2, v_1, and v_2.



■ **Figure 4–24** (a) Beam element subjected to a general load and (b) the statically equivalent nodal force system



■ **Figure 4–25** (a) Beam subjected to a uniformly distributed loading and (b) the equivalent nodal forces to be determined

Example of Load Replacement

To illustrate more clearly the concept of work equivalence, we will now consider a beam subjected to a specified distributed load. Consider the uniformly loaded beam shown in Figure 4–25(a). The support conditions are not shown because they are not relevant to the replacement scheme. By letting $W_{\text{discrete}} = W_{\text{distributed}}$ and by assuming arbitrary ϕ_1, ϕ_2, v_1, and v_2, we will find equivalent nodal forces and moments m_1, m_2, f_{1y}, and f_{2y}. Figure 4–25(b) shows the nodal forces and moments directions as positive based on Figure 4–1.

Using Eqs. (4.4.1) and (4.4.2) for $W_{\text{distributed}} = W_{\text{discrete}}$, we have

$$\int_0^L w(x)v(x)\,dx = m_1\phi_1 + m_2\phi_2 + f_{1y}v_1 + f_{2y}v_2 \tag{4.4.3}$$

where $m_1\phi_1$ and $m_2\phi_2$ are the work due to concentrated nodal moments moving through their respective nodal rotations and $f_{1y}v_1$ and $f_{2y}v_2$ are the work due to the nodal forces moving through nodal displacements. Evaluating the left-hand side of Eq. (4.4.3) by substituting $w(x) = -w$ and $v(x)$ from Eq. (4.1.4), we obtain the work due to the distributed load as

$$\begin{aligned}\int_0^L w(x)v(x)\,dx = &-\frac{Lw}{2}(v_1 - v_2) - \frac{L^2w}{4}(\phi_1 + \phi_2) - Lw(v_2 - v_1) \\ &+\frac{L^2w}{3}(2\phi_1 + \phi_2) - \phi_1\left(\frac{L^2w}{2}\right) - v_1(wL)\end{aligned} \tag{4.4.4}$$

Now using Eqs. (4.4.3) and (4.4.4) for arbitrary nodal displacements, we let $\phi_1 = 1$, $\phi_2 = 0$, $v_1 = 0$, and $v_2 = 0$ and then obtain

$$m_1(1) = -\left(\frac{L^2w}{4} - \frac{2}{3}L^2w + \frac{L^2}{2}w\right) = -\frac{wL^2}{12} \tag{4.4.5}$$

Similarly, letting $\phi_1 = 0, \phi_2 = 1, v_1 = 0$, and $v_2 = 0$ yields

$$m_2(1) = -\left(\frac{L^2 w}{4} - \frac{L^2 w}{3}\right) = \frac{wL^2}{12} \tag{4.4.6}$$

Finally, letting all nodal displacements equal zero except first v_1 and then v_2, we obtain

$$\begin{aligned} f_{1y}(1) &= -\frac{Lw}{2} + Lw - Lw = -\frac{Lw}{2} \\ f_{2y}(1) &= \frac{Lw}{2} - Lw = -\frac{Lw}{2} \end{aligned} \tag{4.4.7}$$

We can conclude that, in general, for any given load function $w(x)$, we can multiply by $v(x)$ and then integrate according to Eq. (4.4.3) to obtain the concentrated nodal forces (and/or moments) used to replace the distributed load. Moreover, we can obtain the load replacement by using the concept of fixed-end reactions from structural analysis theory. Tables of fixed-end reactions have been generated for numerous load cases and can be found in texts on structural analysis such as Reference [2]. A table of equivalent nodal forces has been generated in Appendix D of this text, guided by the fact that fixed-end reaction forces are of opposite sign from those obtained by the work equivalence method.

Hence, if a concentrated load is applied other than at the natural intersection of two elements, we can use the concept of equivalent nodal forces to replace the concentrated load by nodal concentrated values acting at the beam ends, instead of creating a node on the beam at the location where the load is applied. We provide examples of this procedure for handling concentrated loads on elements in beam Example 4.7 and in plane frame Example 5.3.

General Formulation

In general, we can account for distributed loads or concentrated loads acting on beam elements by starting with the following formulation application for a general structure:

$$\{F\} = [K]\{d\} - \{F_0\} \tag{4.4.8}$$

where $\{F\}$ are the concentrated nodal forces and $\{F_0\}$ are called the *equivalent nodal forces*, now expressed in terms of global-coordinate components, which are of such magnitude that they yield the same displacements at the nodes as would the distributed load. Using the table in Appendix D of equivalent nodal forces $\{f_0\}$ expressed in terms of local-coordinate components, we can express $\{F_0\}$ in terms of global-coordinate components.

Recall from Section 3.10 the derivation of the element equations by the principle of minimum potential energy. Starting with Eqs. (3.10.19) and (3.10.20), the minimization of the total potential energy resulted in the same form of equation as Eq. (4.4.8) where $\{F_0\}$ now represents the same work-equivalent force replacement system as given by Eq. (3.10.20a) for surface traction replacement. Also, $\{F\} = \{P\}$, $\{P\}$ [from Eq. (3.10.20)] represents the global nodal concentrated forces. Because we now assume that concentrated nodal forces are not present ($\{F\} = 0$), as we are solving beam problems with distributed loading only in this section, we can write Eq. (4.4.8) as

$$\{F_0\} = [K]\{d\} \tag{4.4.9}$$

On solving for $\{d\}$ in Eq. (4.4.9) and then substituting the global displacements $\{d\}$ and equivalent nodal forces $\{F_0\}$ into Eq. (4.4.8), we obtain the actual global nodal forces $\{F\}$. For example, using the definition of $\{f_0\}$ and Eqs. (4.4.5), (4.4.6), (4.4.7) (or using load case 4 in Appendix D) for a uniformly distributed load w acting over a one-element beam, we have

$$\{F_0\} = \begin{Bmatrix} \dfrac{-wL}{2} \\ \dfrac{-wL^2}{12} \\ \dfrac{-wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} \tag{4.4.10}$$

This concept can be applied on a local basis to obtain the local nodal forces $\{f\}$ in individual elements of structures by applying Eq. (4.4.8) locally as

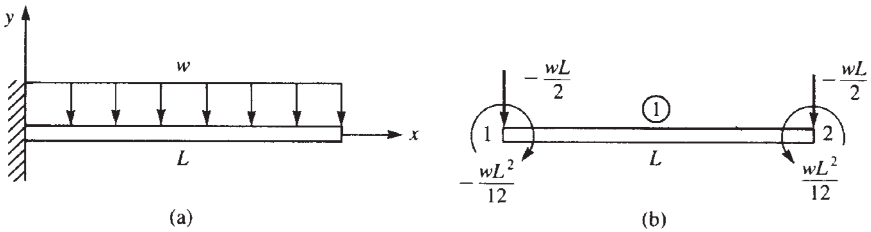
$$\{f\} = [k]\{d\} - \{f_0\} \tag{4.4.11}$$

where $\{f_0\}$ are the equivalent local nodal forces.

Examples 4.6 through 4.8 illustrate the method of equivalent nodal forces for solving beams subjected to distributed and concentrated loadings. We will use global-coordinate notation in Examples 4.6 through 4.8—treating the beam as a general structure rather than as an element.

EXAMPLE 4.6

For the cantilever beam subjected to the uniform load w in Figure 4–26, solve for the right-end vertical displacement and rotation and then for the nodal forces. Assume the beam to have constant EI throughout its length.



■ Figure 4–26 (a) Cantilever beam subjected to a uniformly distributed load and (b) the work equivalent nodal force system

Solution:

We begin by discretizing the beam. Here only one element will be used to represent the whole beam. Next, the distributed load is replaced by its work-equivalent nodal forces as shown in Figure 4–26(b). The work-equivalent nodal forces are those that result from the uniformly distributed load acting over the whole beam given by Eq. (4.4.10). (Or see appropriate load case 4 in Appendix D.) Using Eq. (4.4.9) and the beam element stiffness matrix, we obtain

$$\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ & 4L^2 & -6L^2 & 2L^2 \\ & & 12 & -6L \\ & & & 4L^2 \end{bmatrix}\begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} F_{1y} - \dfrac{wL}{2} \\ M_1 - \dfrac{wL^2}{12} \\ \dfrac{-wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} \tag{4.4.12}$$

where we have applied the work equivalent nodal forces and moments from Figure 4–26(b).

Applying the boundary conditions $v_1 = 0$ and $\phi_1 = 0$ to Eqs (4.4.12) and then partitioning off the third and fourth equations of Eq. (4.4.12), we obtain

$$\frac{EI}{L^3}\begin{bmatrix} 12 & -6L \\ -6L^2 & 4L^2 \end{bmatrix}\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} -\dfrac{wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} \tag{4.4.13}$$

Solving Eq. (4.4.13) for the displacements, we obtain

$$\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \frac{L}{6EI}\begin{bmatrix} 2L^2 & 3L \\ 3L & 6 \end{bmatrix}\begin{Bmatrix} \dfrac{-wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} \tag{4.4.14a}$$

Simplifying Eq. (4.4.14a), we obtain the displacement and rotation as

$$\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{-wL^4}{8EI} \\ \dfrac{-wL^3}{6EI} \end{Bmatrix} \tag{4.4.14b}$$

The negative signs in the answers indicate that v_2 is downward and ϕ_2 is clockwise. In this case, the method of replacing the distributed load by discrete concentrated loads gives exact solutions for the displacement and rotation as could be obtained by classical methods, such as double integration [1]. This is expected, as the work-equivalence method ensures that the nodal displacement and rotation from the finite element method match those from an exact solution.

We will now illustrate the procedure for obtaining the global nodal forces. For convenience, we first define the product $[K]\{d\}$ to be $\{F^{(e)}\}$, where $\{F^{(e)}\}$ are called the *effective global nodal forces*. On using Eq. (4.4.14) for $\{d\}$, we then have

$$\begin{Bmatrix} F_{1y}^{(e)} \\ M_1^{(e)} \\ F_{2y}^{(e)} \\ M_2^{(e)} \end{Bmatrix} = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ \dfrac{-wL^4}{8EI} \\ \dfrac{-wL^3}{6EI} \end{Bmatrix} \tag{4.4.15}$$

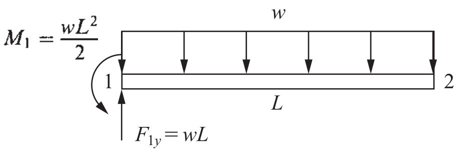
Simplifying Eq. (4.4.15), we obtain

$$\begin{Bmatrix} F_{1y}^{(e)} \\ M_1^{(e)} \\ F_{2y}^{(e)} \\ M_2^{(e)} \end{Bmatrix} = \begin{Bmatrix} \dfrac{wL}{2} \\ \dfrac{5wL^2}{12} \\ \dfrac{-wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} \tag{4.4.16}$$

We then use Eqs. (4.4.10) and (4.4.16) in Eq. (4.4.8) $\{F\} = [K]\{d\} - \{F_0\}$ to obtain the correct global nodal forces as

$$\begin{Bmatrix} F_{1y} \\ M_1 \\ F_{2y} \\ M_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{wL}{2} \\ \dfrac{5wL^2}{12} \\ \dfrac{-wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} - \begin{Bmatrix} \dfrac{-wL}{2} \\ \dfrac{-wL^2}{12} \\ \dfrac{-wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} = \begin{Bmatrix} wL \\ \dfrac{wL^2}{2} \\ 0 \\ 0 \end{Bmatrix} \tag{4.4.17}$$

In Eq. (4.4.17), F_{1y} is the vertical force reaction and M_1 is the moment reaction as applied by the clamped support at node 1. The results for displacement given by Eq. (4.4.14b) and the global nodal forces given by Eq. (4.4.17) are sufficient to complete the solution of the cantilever beam problem.



$$\sum F_y\colon F_{1y} - wL = 0$$

$$\sum M_1\colon \frac{wL^2}{2} - (wL)\frac{L}{2} = 0$$

(c)

■ **Figure 4–26** (c) Free-body diagram and equations of equilibrium for beam of Figure 4–26(a).

A free-body diagram of the beam using the reactions from Eq. (4.4.17) verifies both force and moment equilibrium as shown in Figure 4–26(c).

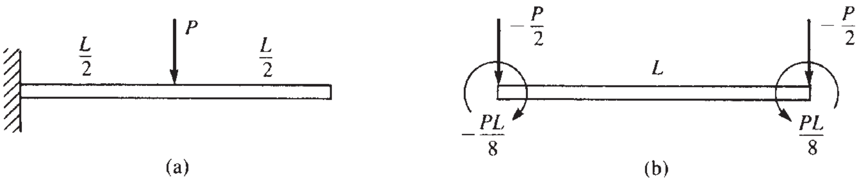
The nodal force and moment reactions obtained by Eq. (4.4.17) illustrate the importance of using Eq. (4.4.8) to obtain the correct global nodal forces and moments. By subtracting the work-equivalent force matrix, $\{F_0\}$, from the product of [K] times $\{d\}$, we obtain the correct reactions at node 1 as can be verified by simple static equilibrium equations. This verification validates the general method as follows:

1. Replace the distributed load by its work-equivalent as shown in Figure 4–26(b) to identify the nodal force and moment used in the solution.
2. Assemble the global force and stiffness matrices and global equations illustrated by Eq. (4.4.12).
3. Apply the boundary conditions to reduce the set of equations as done in previous problems and illustrated by Eq. (4.4.13) where the original four equations have been reduced to two equations to be solved for the unknown displacement and rotation.
4. Solve for the unknown displacement and rotation given by Eq. (4.4.14a) and Eq. (4.4.14b).
5. Use Eq. (4.4.8) as illustrated by Eq. (4.4.17) to obtain the final correct global nodal forces and moments. Those forces and moments at supports, such as the left end of the cantilever in Figure 4–26(a), will be the reactions.

We will solve the following example to illustrate the procedure for handling concentrated loads acting on beam elements at locations other than nodes.

EXAMPLE 4.7

For the cantilever beam subjected to the concentrated load P in Figure 4–27, solve for the right-end vertical displacement and rotation and the nodal forces, including reactions, by replacing the concentrated load with equivalent nodal forces acting at each end of the beam. Assume EI constant throughout the beam.



■ **Figure 4–27** (a) Cantilever beam subjected to a concentrated load and (b) the equivalent nodal force replacement system

Solution:

We begin by discretizing the beam. Here only one element is used with nodes at each end of the beam. We then replace the concentrated load as shown in Figure 4–27(b) by using appropriate loading case 1 in Appendix D. Using Eq. (4.4.9) and the beam element stiffness matrix Eq. (4.1.14), we obtain

$$\frac{EI}{L^3}\begin{bmatrix} 12 & -6L \\ -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{-P}{2} \\ \dfrac{PL}{8} \end{Bmatrix} \tag{4.4.18}$$

where we have applied the nodal forces from Figure 4–27(b) and the boundary conditions $v_1 = 0$ and $\phi_1 = 0$ to reduce the number of matrix equations for the usual longhand solution. Solving Eq. (4.4.18) for the displacements, we obtain

$$\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \frac{L}{6EI}\begin{bmatrix} 2L^2 & 3L \\ 3L & 6 \end{bmatrix}\begin{Bmatrix} \dfrac{-P}{2} \\ \dfrac{PL}{8} \end{Bmatrix} \tag{4.4.19}$$

Simplifying Eq. (4.4.19), we obtain the displacement and rotation as

$$\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{-5PL^3}{48EI} \\ \dfrac{-PL^2}{8EI} \end{Bmatrix} \begin{matrix} \downarrow \\ \curvearrowright \end{matrix} \tag{4.4.20}$$

To obtain the unknown nodal forces, we begin by evaluating the effective nodal forces $\{F^{(e)}\} = [K]\{d\}$ as

$$\begin{Bmatrix} F_{1y}^{(e)} \\ M_1^{(e)} \\ F_{2y}^{(e)} \\ M_2^{(e)} \end{Bmatrix} = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ \dfrac{-5PL^3}{48EI} \\ \dfrac{-PL^2}{8EI} \end{Bmatrix} \tag{4.4.21}$$

Simplifying Eq. (4.4.21), we obtain

$$\begin{Bmatrix} F_{1y}^{(e)} \\ M_1^{(e)} \\ F_{2y}^{(e)} \\ M_2^{(e)} \end{Bmatrix} = \begin{Bmatrix} \dfrac{P}{2} \\ \dfrac{3PL}{8} \\ \dfrac{-P}{2} \\ \dfrac{PL}{8} \end{Bmatrix} \tag{4.4.22}$$

Then using Eq. (4.4.22) and the equivalent nodal forces from Figure 4–27(b) in Eq. (4.4.8), we obtain the correct nodal forces as

$$\begin{Bmatrix} F_{1y} \\ M_1 \\ F_{2y} \\ M_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{P}{2} \\ \dfrac{3PL}{8} \\ \dfrac{-P}{2} \\ \dfrac{PL}{8} \end{Bmatrix} - \begin{Bmatrix} \dfrac{-P}{2} \\ \dfrac{-PL}{8} \\ \dfrac{-P}{2} \\ \dfrac{PL}{8} \end{Bmatrix} = \begin{Bmatrix} P \\ \dfrac{PL}{2} \\ 0 \\ 0 \end{Bmatrix} \tag{4.4.23}$$

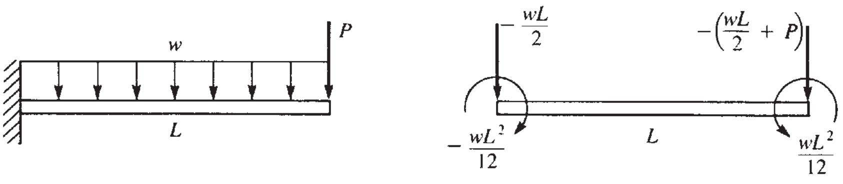
We can see from Eq. (4.4.23) that F_{1y} is equivalent to the vertical reaction force and M_1 is the reaction moment as applied by the clamped support at node 1.

Again, the reactions obtained by Eq. (4.4.23) can be verified to be correct by using static equilibrium equations to validate once more the correctness of the general formulation and procedures summarized in the steps given after Example 4.6.

To illustrate the procedure for handling concentrated nodal forces and distributed loads acting simultaneously on beam elements, we will solve the following example.

EXAMPLE 4.8

For the cantilever beam subjected to the concentrated free-end load P and the uniformly distributed load w acting over the whole beam as shown in Figure 4–28, determine the free-end displacements and the nodal forces.



■ **Figure 4–28** (a) Cantilever beam subjected to a concentrated load and a distributed load and (b) the equivalent nodal force replacement system

Solution:

Once again, the beam is modeled using one element with nodes 1 and 2, and the distributed load is replaced as shown in Figure 4–28(b) using appropriate loading case 4 in Appendix D. Using the beam element stiffness Eq. (4.1.14), we obtain

$$\frac{EI}{L^3}\begin{bmatrix} 12 & -6L \\ -6L & 4L^2 \end{bmatrix}\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{-wL}{2} - P \\ \dfrac{wL^2}{12} \end{Bmatrix} \tag{4.4.24}$$

where we have applied the nodal forces from Figure 4–28(b) and the boundary conditions $v_1 = 0$ and $\phi_1 = 0$ to reduce the number of matrix equations for the usual longhand solution. Solving Eq. (4.4.24) for the displacements, we obtain

$$\begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{-wL^4}{8EI} - \dfrac{PL^3}{3EI} \\ \dfrac{-wL^3}{6EI} - \dfrac{PL^2}{2EI} \end{Bmatrix} \begin{matrix} \downarrow \\ \curvearrowright \end{matrix} \tag{4.4.25}$$

Next, we obtain the effective nodal forces using $\{F^{(e)}\} = [K]\{d\}$ as

$$\begin{Bmatrix} F_{1y}^{(e)} \\ M_1^{(e)} \\ F_{2y}^{(e)} \\ M_2^{(e)} \end{Bmatrix} = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ \dfrac{-wL^4}{8EI} - \dfrac{PL^3}{3EI} \\ \dfrac{-wL^3}{6EI} - \dfrac{PL^2}{2EI} \end{Bmatrix} \tag{4.4.26}$$

Simplifying Eq. (4.4.26), we obtain

$$\begin{Bmatrix} F_{1y}^{(e)} \\ M_1^{(e)} \\ F_{2y}^{(e)} \\ M_2^{(e)} \end{Bmatrix} = \begin{Bmatrix} P + \dfrac{wL}{2} \\ PL + \dfrac{5wL^2}{12} \\ -P - \dfrac{wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} \tag{4.4.27}$$

Finally, subtracting the equivalent nodal force matrix [see Figure 4–27(b)] from the effective force matrix of Eq. (4.4.27), we obtain the correct nodal forces as

$$\begin{Bmatrix} F_{1y} \\ M_1 \\ F_{2y} \\ M_2 \end{Bmatrix} = \begin{Bmatrix} P + \dfrac{wL}{2} \\ PL + \dfrac{5wL^2}{12} \\ -P - \dfrac{wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} - \begin{Bmatrix} \dfrac{-wL}{2} \\ \dfrac{-wL^2}{12} \\ \dfrac{-wL}{2} \\ \dfrac{wL^2}{12} \end{Bmatrix} = \begin{Bmatrix} P + wL \\ PL + \dfrac{wL^2}{2} \\ -P \\ 0 \end{Bmatrix} \tag{4.4.28}$$

From Eq. (4.4.28), we see that F_{1y} is equivalent to the vertical reaction force, M_1 is the reaction moment at node 1, and F_{2y} is equal to the applied downward force P at node 2. [Remember that only the equivalent nodal force matrix is subtracted, not the original concentrated load matrix. This is based on the general formulation, Eq. (4.4.8).]

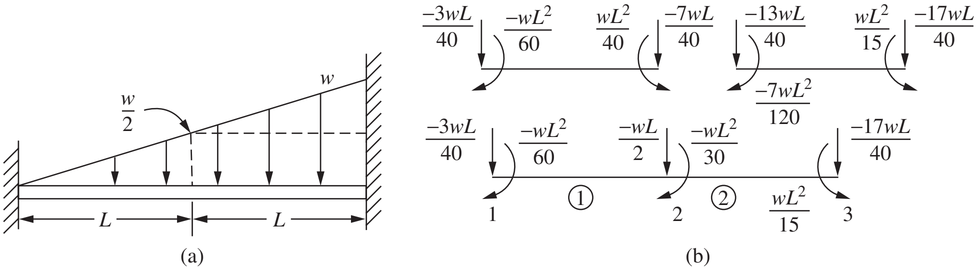
To generalize the work-equivalent method, we apply it to a beam with more than one element as shown in the following Example 4.9.

EXAMPLE 4.9

For the fixed–fixed beam subjected to the linear varying distributed loading acting over the whole beam shown in Figure 4–29(a) determine the displacement and rotation at the center and the reactions.

Solution:

The beam is now modeled using two elements with nodes 1, 2, and 3 and the distributed load is replaced as shown in Figure 4–29(b) using the appropriate load cases 4 and 5 in Appendix D. Note that load case 5 is used for element one as it has only the linear varying distributed load acting on it with a high end value of $w/2$ as shown in Figure 4–29(a), while both load cases 4 and 5 are used for element two as the distributed load is divided into a uniform part with magnitude $w/2$ and a linear varying part with magnitude at the high end of the load equal to $w/2$ also.



■ **Figure 4–29** (a) Fixed-fixed beam subjected to linear varying line load and (b) the equivalent nodal force replacement system

Using the beam element stiffness Eq. (4.1.14) for each element, we obtain

$$[k^{(1)}]=\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \quad [k^{(2)}] = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \tag{4.4.28}$$

The boundary conditions are $v_1 = 0$, $\phi_1 = 0$, $v_3 = 0$, and $\phi_3 = 0$. Using the direct stiffness method and Eqs. (4.4.28) to assemble the global stiffness matrix, and applying the boundary conditions, we obtain

$$\begin{Bmatrix} F_{2y} \\ M_2 \end{Bmatrix} = \begin{Bmatrix} \dfrac{-wL}{2} \\ \dfrac{-wL^2}{20} \end{Bmatrix} = \frac{EI}{L^3}\begin{bmatrix} 24 & 0 \\ 0 & 8L^2 \end{bmatrix} = \begin{Bmatrix} v_2 \\ \phi_2 \end{Bmatrix} \tag{4.4.29}$$

Solving Eq. (4.4.29) for the displacement and slope, we obtain

$$v_2 = \frac{-wL^4}{48EI} \qquad \phi_2 = \frac{-wL^3}{240EI} \tag{4.4.30}$$

Next, we obtain the effective nodal forces using $\{F^{(e)}\} = [K]\{d\}$ as

$$\begin{Bmatrix} F_{1y}^{(e)} \\ M_1^{(e)} \\ F_{2y}^{(e)} \\ M_2^{(e)} \\ F_{3y}^{(e)} \\ M_3^{(e)} \end{Bmatrix} = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L & 0 & 0 & 0 \\ 6L & 4L^2 & -6L & 2L^2 & 0 & 0 & 0 \\ -12 & -6L & 24 & 0 & -12 & -12 & 6L \\ 6L & 2L^2 & 0 & 8L^2 & -6L & -6L & 2L^2 \\ 0 & 0 & -12 & -6L & 12 & 12 & -6L \\ 0 & 0 & 6L & 2L^2 & 6L & -16L & 4L^2 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ \dfrac{-wL^4}{48EI} \\ \dfrac{-wL^3}{240EI} \\ 0 \\ 0 \end{Bmatrix} \tag{4.4.31}$$

Solving for the effective forces in Eq. (4.4.31), we obtain

$$\begin{aligned} F_{1y}^{(e)} &= \frac{9wL}{40} & M_1^{(e)} &= \frac{7wL^2}{60} \\ F_{2y}^{(e)} &= \frac{-wL}{2} & M_2^{(e)} &= \frac{-wL^2}{30} \\ F_{3y}^{(e)} &= \frac{11wL}{40} & M_3^{(e)} &= \frac{-2wL^2}{15} \end{aligned} \tag{4.4.32}$$

Finally, using Eq. (4.4.8) we subtract the equivalent nodal force matrix based on the equivalent load replacement shown in Figure 4–29(b) from the effective force matrix given by the results in Eq. (4.4.32), to obtain the correct nodal forces and moments as

$$\begin{Bmatrix} F_{1y} \\ M_1 \\ F_{2y} \\ M_2 \\ F_{3y} \\ M_3 \end{Bmatrix} = \begin{Bmatrix} \dfrac{9wL}{40} \\ \dfrac{7wL^2}{60} \\ \dfrac{-wL}{2} \\ \dfrac{-wL^2}{30} \\ \dfrac{11wL}{40} \\ \dfrac{-2wL^2}{15} \end{Bmatrix} - \begin{Bmatrix} \dfrac{-3wL}{40} \\ \dfrac{-wL^2}{60} \\ \dfrac{-wL}{2} \\ \dfrac{-wL^2}{30} \\ \dfrac{-17wL}{40} \\ \dfrac{wL^2}{15} \end{Bmatrix} = \begin{Bmatrix} \dfrac{12wL}{40} \\ \dfrac{8wL^2}{60} \\ 0 \\ 0 \\ \dfrac{28wL}{40} \\ \dfrac{-3wL^2}{15} \end{Bmatrix} \tag{4.4.33}$$